Browser-Based Game for Teaching Cyber Security Concepts

Game mechanics, Educational objectives and User flow

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ISG Final Year Project

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1 - Game description

In a distant future, the world of Cipheria stands on the edge of destruction. Once a harmonious realm powered by the Motherboard—a living network that sustains life and preserves the memories of every citizen—it now descends into chaos as powerful forces exploit the cracks in its defences, corrupting the Motherboard and twisting the very world it was created to protect.

The player takes on the role of a young apprentice guardian, tasked with defending Cipheria and defeating its enemies alongside one of Cipheria's legendary elders.

Through this journey, the player will learn the art of cybersecurity through story-driven quests, immersive puzzles, and strategic mini-games. Each quest transforms a real-world cybersecurity concept into an interactive challenge—teaching players how to protect systems, identify risks, and outsmart attackers while restoring balance to Cipheria.

2 - Educational objectives

The goal of the game is to teach foundational cybersecurity principles through interactive, story-driven gameplay. Each quest transforms a real-world cybersecurity concept into a challenge using metaphors, allowing players to learn by doing rather than learning passively.

By completing all the quests, players will have an understanding of the following cybersecurity concepts:

1. What is cybersecurity - understanding the importance of protecting digital assets
2. Understanding cyber risk - identifying vulnerabilities, threats, exploits and the impact each risk has
3. Common cyber attacks - recognising each attack and its countermeasures
4. CIA triad
5. IAAA principles
6. People and security - recognising social engineering and practising strong password hygiene
7. Defensive mechanisms - understanding firewalls, antivirus systems, encryption and intrusion detection
8. Operating systems - learning how different systems have their own vulnerabilities
9. Security controls - distinguishing between technical, administrative and physical controls
10. Cryptography basics

Each concept will be introduced progressively, reinforced through challenges and consolidated through boss encounters, which will test the player's comprehension.

The game will follow a structured educational loop designed to introduce, reinforce and assess the player's learning throughout the gameplay.

- ➢ Introduction phase - concept introduced via story dialogue and a simple quest. The player will learn through guided tutorials or easy challenges.
- ➢ Consolidation phase - concepts are reinforced through progressively harder puzzles.
- ➢ Final boss phase - Each concept will have a boss challenge, which will test the mastery of that concept. The player's understanding is tested.

3 - Visual style and theme

The core gameplay is shown in 2D pixel art, using both top-down isometric and side-scrolling views to make the game more engaging and to separate different types of quests. Both views keep the same cyberpunk pixel art style, giving the game a consistent look and feel.

4 - Quest descriptions

4.1 What is cybersecurity

Quest - The player is welcomed to Cipheria, a futuristic world powered by a massive Tower where the Motherboard resides. The Motherboard is the heart of Cipheria - a powerful source made up of three crystals that keep the world alive and connected. But dark forces seek to steal the crystals, for whoever controls the Motherboard controls the entire world.

In this level, the player must climb the Tower to reach the Motherboard at its peak, defending it from waves of enemy robots and drones trying to break in and steal its power.

Teaching - Cybersecurity = protecting valuable assets from threats.

4.2 Understanding cyber risk

Quest - After reaching the top of the Tower, the player enters the Motherboard, the heart of Cipheria. Inside, the player must explore its inner systems to locate and repair areas of damage before they worsen.

- **Vulnerabilities:** Cracks or damage within the Motherboard's structure.
- **Exploits:** Enemies squeezing through those cracks to cause more harm.
- **Threats:** The enemies themselves attacking the system.
- **Threat Actor:** A boss enemy commanding and coordinating the attacks.

As the player navigates the level, a risk percentage appears beside each damaged area, showing the chance of an enemy breaking through if it isn't repaired in time.

At the end of the quest, the player exits the Tower and speaks with Cipheria's citizens, witnessing firsthand how the attacks have affected their world - linking the player's actions to real-world consequences.

Teaching - Understand the meaning of each of the following terms: vulnerabilities, exploits, threats, threat actor, risk and impact.

4.3 Common cyber attacks

Quest - After protecting the Motherboard, the Elder sends the player to the Null Domain - a dangerous region where chaos and corruption reign. The Elder explains that to become a true Guardian, the apprentice must learn to recognize and defend against the most common cyber threats that endanger Cipheria.

Inside the Null Domain, the player faces a series of encounters that represent real-world cyberattacks:

- **Phishing:** A trickster NPC disguises themselves as the Elder and tries to deceive the player into giving away the keys to the Tower to gain access to the Motherboard.

- **Malware:** Swarms of slime-like creatures that multiply each time they are touched, spreading digital corruption through the area.

- **DDoS (Distributed Denial of Service):** Endless waves of small drones that overwhelm the player's defenses through sheer numbers.

- **Ransomware:** A cyborg enemy steals and locks away the player's sword, forcing them to either solve a puzzle or surrender the keys to the Motherboard.

The player must face each type of attack three times, with each encounter changing its format or appearance to test the player's ability to recognize threats in different forms.
In the final stage, the player will need to match each cyberattack name to the threat in front of them, reinforcing pattern recognition and situational awareness.

Teaching - Identify common cyberattacks and their countermeasures, emphasizing that these threats can appear in different forms and must be recognized through their behavior and intent, not just their appearance.

4.4 CIA

Quest - After returning from the Null Domain, the player arrives back at the main tower to find that the 3 crystals that power the Motherboard have been stolen and tampered with. The Elder explains that these crystals represent the CIA - Confidentiality, Integrity and Availability - and that they are essential to keep the balance in Cipheria. The player must locate and restore each crystal:

- **Confidentiality crystal** - hidden deep within a dark alley of Cipheria's lower districts. To find, the player must earn the trust of local citizens who may have witnessed anything. The player will engage in dialogue puzzles, choosing responses to build trust and demonstrate discretion. Fake informants attempt to trick the player into revealing too much information or sharing secrets, enforcing that not everyone can be trusted. The player must use a passkey to prove their identity before the citizens reveal their clues.
- **Integrity crystal** - still located within the Motherboard, but has been tampered with. The player must analyze it and restore it to its original state. If the player fails to fully repair it, the virus begins to spread, infecting the Motherboard and as a consequence Cipheria
- **Availability crystal** - Trapped inside a chamber blocked by endless waves of drones. The player must restore access by clearing the enemies and placing it back on its pedestal to make it available again.

Teaching - Principles of the CIA triad and that protecting all 3 ensures the stability and security of a system.

4.5 IAAA

Quest - After heroically restoring balance to Cipheria, the Rulers of Cipheria invite the player to their fortress to honor their achievements. Enclosed by three walls, the fortress represents layers of security.

Before entering, the player receives a Cyber Seal along with an official invitation, both of which will be used throughout the entry process. To gain access, the player must successfully pass through each security layer, each representing a stage of the IAAA framework:

- **Identification (First Wall):**
  Upon arriving at the outer gate, the guards stop the player and ask for their name. They check it against the official visitor list to confirm the player's identity.
- **Authentication (Second Wall):**
  Once past the first wall, the player must present their Cyber Seal - a unique, encrypted symbol given in the invitation. Guards verify that the seal matches the record on the visitor list. Only if it matches will the player be allowed to proceed.
- **Authorization (Third Wall):**
  At the innermost gate, the player is asked to show proof of their invitation, confirming they are not only identified and authenticated but also authorized to enter the fortress and meet the rulers.

After successfully passing all three walls, the player is assigned a Scribe, that records all actions during their stay inside the fortress — representing the final principle: **Accountability**.

Teaching - IAAA principles, their importance and how each layer works together to secure access, verify identity and ensure each action is accountable within a secure system.

4.6 People and security

Quest - After having met the rulers of Cipheria, the player makes their way back to the main tower. On their journey they will interact with citizens, who seek guidance and protection from the newly arising threats. However, not everyone is who they say they are. Some citizens are tricksters in disguise, using deception and persuasion to try convince the player to reveal some of the citizens' passkeys or the player's cyber seal. Other citizens will genuinely ask for help, explaining that they need a guardian to create strong passkeys for their homes. The player must craft secure passwords for them.

Teaching - Sometimes people are often the weakest link in cybersecurity and that being aware of social engineering and reckognising it is crucial.

4.7 Defensive mechanisms

Quest - As the player returns to the main tower, the Elder tasks them with building and strengthening the tower's defences to protect the Motherboard from future attacks. The player must craft the following defences:

- Firewall - a barrier surrounding the tower blocking enemy drones or robots allowing only trusted entities to enter.
- Antivirus - a maintenance robot which roams the tower's interior cleansing any viruses or corruption.
- Encryption - The door to the Motherboard will only show itself if a special sequence of characters is entered.
- IDS - a radio like creature, which monitors for suspect behaviour and announces if enemies manage to sneak in.

Teaching - The importance of defensive mechanisms and that placing multiple layers of protection is good practice for maintaining long-term protection.

4.8 Operating systems

Quest - Cipheria is made up of several kingdoms, each ruled by a unique operating system and its own digital language:

- **The Purple Kingdom – Linux**
- **The Green Kingdom – Windows**
- **The Yellow Kingdom – Mac**

An enemy hacker sends the player a series of encrypted puzzles, each written in one of these kingdom's languages. To decode them, the player must travel to the corresponding kingdom, where the language becomes readable.

In each kingdom, the player must translate and solve puzzles that reveal fragments of the hacker's plan. Once all puzzles are solved, the player can trace the source of the attack and catch one of Cipheria's evil masterminds.

Teaching - Each operating system have unique structures, languages and vulnerabilities and how it is important to understand how each system differs.

4.9 Security controls

Quest - After successfully completing every challenge and proving their heroic nature, the Elder assigns the player one final test before officially crowning them as a Guardian of Cipheria. The player must design and build a new Abbey for the Elders, implementing three key categories of security controls to protect it from any attacks.

- Technical controls: construct walls, moats, and defense towers to protect the Abbey's systems and prevent unauthorized access.
- Administrative controls: create a set of rules and policies for those who live and work within the Abbey.
- Physical controls: hire guards and install locks on gates and doors, representing physical barriers that protect against any intrusion.

Once the construction is complete, the Abbey's security will be tested in a mock attack simulation. If the player's security design is strong, the Abbey will pass the test. If too few controls are in place, the player must reattempt until the Abbey passes the test.

Teaching - How each category of security controls work together to create a resilient defence system.

4.10 Cryptography basics

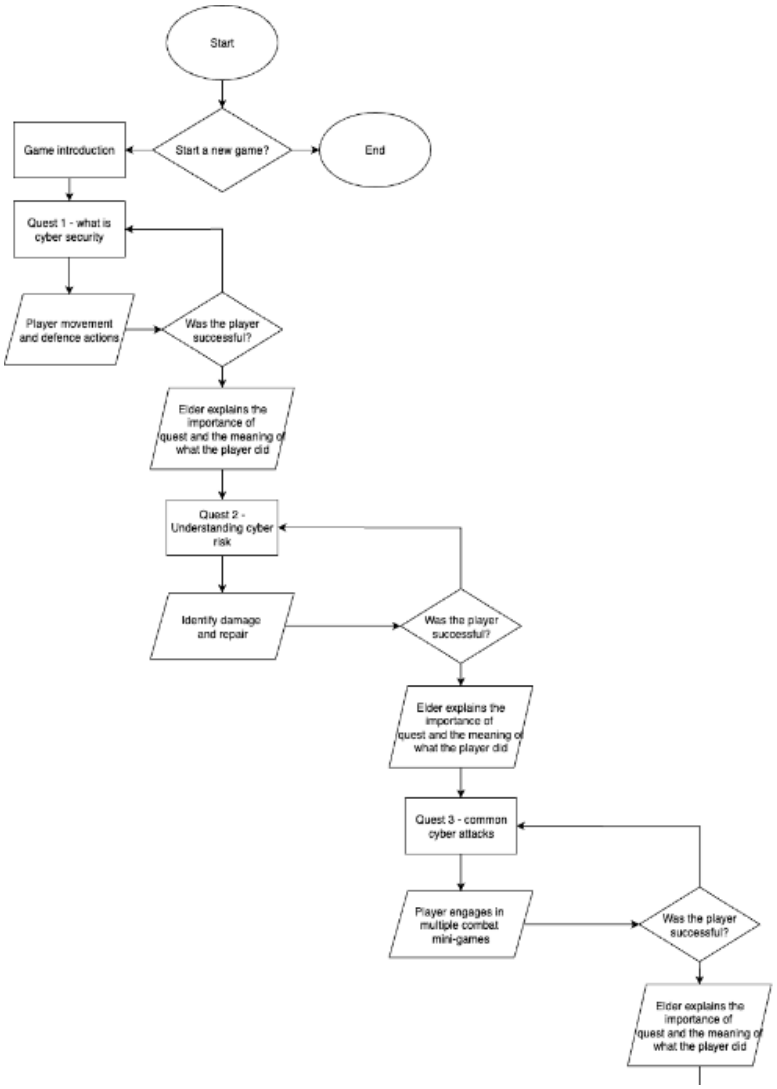
Quest - A mysterious message appears on the Motherboard - a distress signal written in a way which is unreadable. The Elder believes it could come from one of the missing guardians who travelled into the Null Domain to try defeat some of Cipheria's worst criminals. The player must decode the message and reveal its meaning. Once decrypted, the message confirms the guardian is still alive and in danger. To communicate, the player must encrypt new messages using the same cipher. This starts a coded conversation between the player and missing guardian, with each exchange revealing more about:

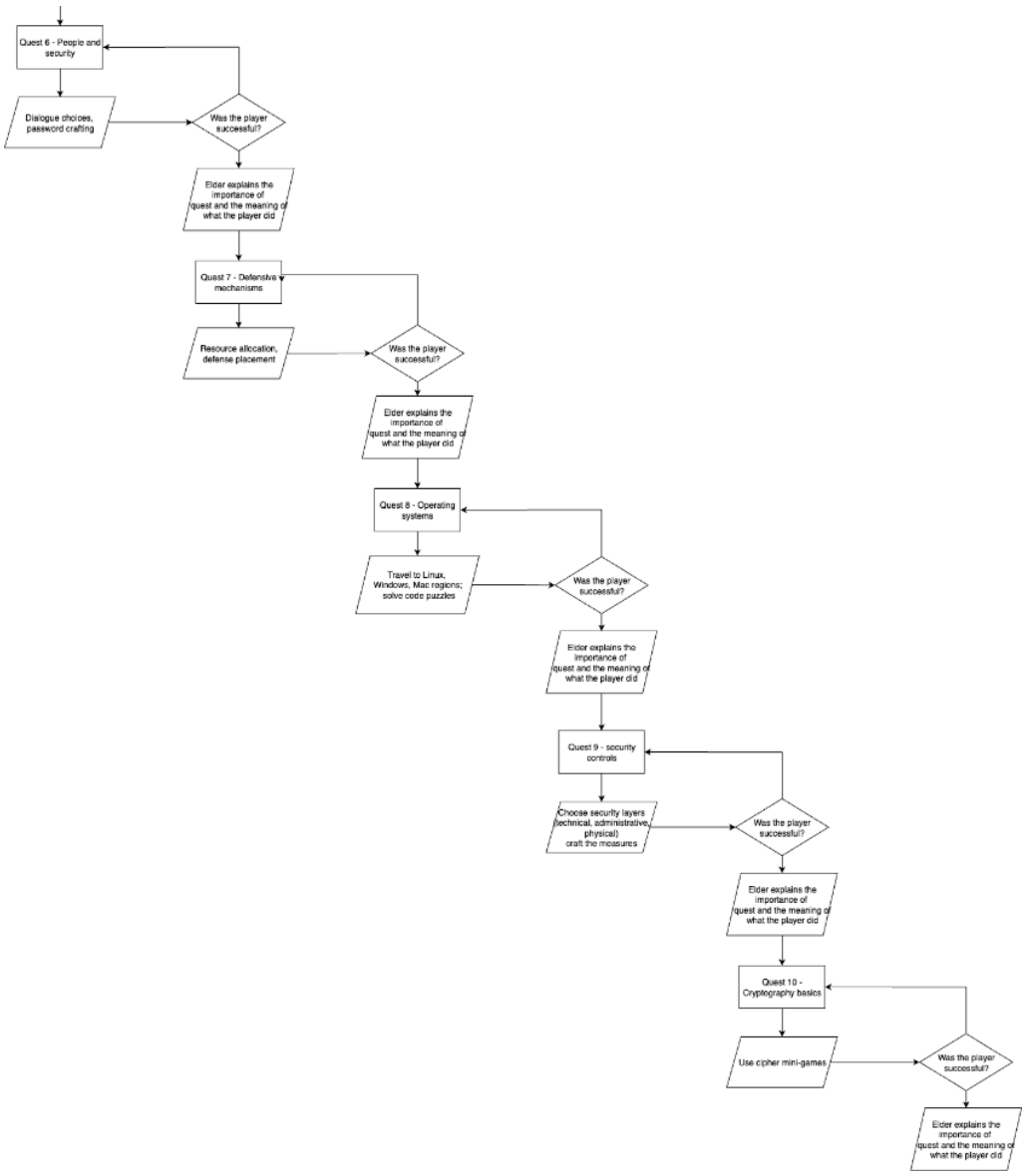
- What happened during the guardian's failed mission in the Null Domain
- The identities of the final bosses
- The location of the missing guardia

Each message becomes more complex to decode, and different cyphers may be used to ensure that the conversation is secure and confidential.

Teaching - Foundations of cryptography, including encryption, decryption and secure communication.

5 - User flow




Quest 6 - People and security
Dialogue choices, password crafting
Was the player successful?
Elder explains the importance of quest and the meaning of what the player did
Quest 7 - Defensive mechanisms
Resource allocation, defense placement
Was the player successful?
Elder explains the importance of quest and the meaning of what the player did
Quest 8 - Operating systems
Travel to Linux, Windows, Mac regions; solve code puzzles
Was the player successful?
Elder explains the importance of quest and the meaning of what the player did
Quest 9 - security controls
Choose security layers (technical, administrative, physical) craft the measures
Was the player successful?
Elder explains the importance of quest and the meaning of what the player did
Quest 10 - Cryptography basics
Use cipher mini-games
Was the player successful?
Elder explains the importance of quest and the meaning of what the player did